

AMRIT RAJ

+91-8271425886 | amritrajsingh179@gmail.com | [GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [GeeksforGeeks](#) | [Hashnode](#) | Lucknow, India

Experience

Pinnacle Labs

Web Development Intern

Remote

June – July 2026

- Engineered a full-stack **event management platform** on **Next.js**, and **TypeScript**, enabling organizers to create, publish, and manage events end-to-end
- Resolved booking race conditions using **row-level locking**, and implemented **idempotent APIs** to prevent redundant calls, ensuring reliable event registrations under concurrent requests
- Designed and implemented **RESTful APIs** with **Drizzle ORM**, enforcing atomic transactions for reliable data integrity
- Automated **email ticket and reminder delivery** on event creation, streamlining outreach and eliminating manual invite tracking

Technical Skills

Languages:	JavaScript, TypeScript, Java
Backend:	Node.js, Express.js, REST APIs, JWT, WebSockets
AI/ML:	LangChain, RAG, Vector Databases, OpenRouter, Inngest
Databases & ORM:	PostgreSQL, Drizzle, Redis, Pinecone
Frontend:	React.js, Next.js, HTML, CSS
Infrastructure:	Docker
Developer Tools:	Git, GitHub, Postman
Core Subjects:	DSA, OS, DBMS, OOP, Computer Networks

Projects

PRRabbit – AI-Powered GitHub PR Reviewer

[Live](#) [GitHub](#)

- Architected an **automated PR review system** as a GitHub App, using **Inngest** for async job orchestration to decouple webhook response times from long-running AI inference
- Implemented **diff parsing with line-level mapping**, translating raw unified diffs into addressable code positions for accurate inline commenting via the **GitHub Reviews API**
- Integrated **Pinecone-backed RAG retrieval** to pull relevant repository context into each review prompt, grounding feedback in the actual codebase instead of the diff alone
- Abstracted model access through **OpenRouter**, decoupling the review pipeline from any single LLM provider

Prepr AI – Real-Time AI Mock Interview Platform

[Live](#) [GitHub](#)

- Engineered a real-time **AI voice interview platform** using OpenAI's **Realtime WebRTC API (gpt-realtime-2.1-mini)**, enabling low-latency bidirectional audio streaming for live mock interviews
- Built dynamic, context-aware question generation by parsing candidate **resumes (PDF)**, job descriptions, and **GitHub repositories** to tailor interviews to individual profiles
- Designed a **roleplay-based prompting system** with chain-of-thought reasoning and strict scope guardrails, keeping the AI interviewer in-persona and preventing off-task or out-of-scope responses

Muzzix – Collaborative Music Queue App

[Live](#) [GitHub](#)

- Built a real-time collaborative music queue app where users join shared rooms, add songs, upvote songs, and vote to control playback order collectively synced live across all connected clients
- Eliminated audio drift via **NTP-style clock sync**, keeping playback skew under 500ms across all connected clients
- Reduced vote queue processing from **$O(N \log N)$ to $O(\log N)$** by switching to Redis Sorted Sets, removing per-vote server recalculation
- Achieved **sub-20ms event propagation** across scaled WebSocket servers using Redis Pub/Sub broadcast architecture
- Resolved concurrent vote race conditions with **Redis atomic operations**, guaranteeing correct counts under simultaneous writes with no locking overhead

Achievements

- Solved 800+ DSA problems** across LeetCode and GeeksforGeeks, maintaining consistent active coding practice
- Currently **serving as Webmaster** for the IEEE Student Branch at AIMT, managing and maintaining the official website

Education

Ambalika Institute of Management and Technology

B.Tech in Computer Science and Engineering

Lucknow, India

2023 – 2027